

For each weather–time group:

Historical observations

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Calculate mean price → $P_{\text{base}}(W,T)$

Calculate mean distance → $D_{\text{base}}(W,T)$

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For every historical observation:

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$\Delta P = (P - P_{\text{base}}) / P_{\text{base}}$

$\Delta D = (D - D_{\text{base}}) / D_{\text{base}}$

↓

$\epsilon = |\Delta P / \Delta D|$

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Calculate:

$\mu_{\epsilon}(W,T)$

$\sigma_{\epsilon}(W,T)$

↓

Generate simulated customer sensitivity

$\epsilon(0,W,T) > 0$

So my recommendation is:

$$P_{W,T}^{base} = \text{mean price}$$

$$D_{W,T}^{base} = \text{mean distance}$$

and:

$$\epsilon_i = \left| \frac{(P_i - P^{base})/P^{base}}{(D_i - D^{base})/D^{base}} \right|$$

Then compute the mean and standard deviation from these **positive historical sensitivity magnitudes**.